

# An explicit family of classical states refuting a conjecture of Linden and Winter

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## Abstract

Linden and Winter (*Commun. Math. Phys.* 259:129–138, 2005) proved that any four-party quantum state with  $I(A ; C|B) = I(C ; B|A) = I(A ; B|D) = 0$  satisfies  $I(C ; D) \geq I(C ; AB)$ , and conjectured (their Conjecture 6) that this constrained inequality is stable: that positive constants  $k_1, k_2, k_3$  exist with  $k_1 I(A ; C|B) + k_2 I(C ; B|A) + k_3 I(A ; B|D) + I(C ; D) - I(C ; AB) \geq 0$  for all quadripartite states. We exhibit an explicit one-parameter family of *classical* probability distributions on four bits — six atoms, integer weights in closed form — on which  $I(A ; C|B)$  and  $I(A ; B|D)$  vanish identically while the ratio  $[I(C ; AB) - I(C ; D)]/I(C ; B|A)$  grows without bound. Consequently no choice of positive constants satisfies the conjectured inequality, even for classical states: the conjecture is false. The vanishing of the two conditional mutual informations is an exact combinatorial identity; the divergence of the ratio follows from an elementary asymptotic analysis, and is independently certified by directed-rounding interval arithmetic, which also yields explicit refuting states for every  $(k_1, k_2, k_3)$  with  $k_2 \leq 10^{24}$ . All certificates re-verify from a single dependency-free Python script.

## 1 Introduction

Which linear inequalities constrain the von Neumann entropies of the reduced states of a multipartite quantum state? For up to three parties the answer is classical: strong subadditivity and weak monotonicity [1] generate all inequalities, and Pippenger [2] asked whether anything new appears for four or more parties. No unconstrained inequality beyond the Lieb–Ruskai family has been found since 1973.

Linden and Winter [3] made the first advance past this point, proving a new *constrained* inequality:

**Theorem (Linden–Winter).** If a four-party state  $\rho_{ABCD}$  satisfies  $I(A ; C|B) = I(C ; B|A) = I(A ; B|D) = 0$ , then  $I(C ; D) \geq I(C ; AB)$ .

They closed the paper by conjecturing that the constraint can be relaxed linearly:

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**Conjecture 6 (Linden–Winter).** There exist positive real constants  $k_1, k_2, k_3$  such that for all quadripartite quantum states,

$$k_1 I(A ; C|B) + k_2 I(C ; B|A) + k_3 I(A ; B|D) + I(C ; D) - I(C ; AB) \geq 0. \quad (1)$$

A proof would have produced the first new unconstrained inequality for the von Neumann entropy in over five decades. We show the conjecture fails, already for classical states.

**Theorem 1.** *No positive constants  $k_1, k_2, k_3$  satisfy (1). Specifically, for the family  $W_L$  of classical distributions in Definition 3:  $I(A ; C|B)$  and  $I(A ; B|D)$  vanish identically on the family, and  $R_L := [I(C ; AB) - I(C ; D)]/I(C ; B|A) \rightarrow \infty$  as  $L \rightarrow \infty$ . Hence for any positive  $(k_1, k_2, k_3)$ , every  $L$  with  $R_L > k_2$  gives  $f_k(W_L) < 0$  in (1).*

**Corollary 2.** *The Shannon-entropy version of the Linden–Winter constrained inequality is essentially conditional in the sense of Kaced–Romashchenko: no positive constants make (1) hold even for all classical distributions.*

*Proof.* The family of Theorem 1 is classical. □

Classical distributions embed as diagonal quantum states, so a classical counterexample refutes the quantum conjecture. The constrained theorem itself is untouched: on the family, two of its three hypotheses hold exactly but the third,  $I(C ; B|A) = 0$ , fails — by an amount that shrinks too fast, which is precisely what the conjecture forbade. In the vocabulary of Kaced and Romashchenko [7], the Linden–Winter constrained inequality is *essentially conditional*: no finite Lagrange multipliers lift it to an unconstrained inequality. They established this phenomenon for five conditional *Shannon* inequalities, including the Zhang–Yeung 1997 conditional inequality; the present family shows the same fate for the one known conditional *von Neumann* inequality, and since the family is classical, it simultaneously extends their classical list: the Shannon version of the Linden–Winter inequality is essentially conditional as well (Corollary 2).

Two prior results frame what is and is not being shown. Ibinson, Linden and Winter [4] proved that the robust-structure heuristic behind the conjecture fails — tripartite quantum states can be far from Markov chains while  $I(A ; C|B)$  is small — and wrote that this “casts doubts” on the proof route; but their examples are tripartite, concern Markov-chain distance rather than the quadripartite functional in (1), and are essentially quantum (the classical ratio in their setting is bounded). Killing the heuristic left the conjecture itself open; the present family closes it, classically. In the positive direction, Majenz [8, Thm. 2.9] proved that on the slice  $I(A ; C|B) = I(B ; C|A) = 0$  the choice  $\kappa_3 = 1$  already works. Our family lives on the *complementary* slice: it zeroes  $I(A ; C|B)$  and  $I(A ; B|D)$  while keeping  $I(C ; B|A) > 0$  (certified strictly positive in Section 4) — exactly the term his theorem does not cover, so the two results are consistent and jointly pin the failure to the  $\kappa_2$  direction.

**Provenance.** The family was found by an automated research system conducting a systematic search over supports and weight patterns, and the numerical certificates below were produced and cross-checked by four independently written implementations (no shared code), including the dependency-free verifier published with this note. The mathematics presented here — the identities of Lemma 4 and the asymptotics of Proposition 5 — is human-checkable by hand.

## 2 The family

**Proposition 3** (Definition). *For an integer  $L \geq 2$  let  $W_L$  be the probability distribution on  $\{0, 1\}^4$  (bits ordered  $A, B, C, D$ ) supported on six atoms, with weights proportional to:*

| <i>atom</i> ( $A, B, C, D$ ) | <i>weight</i>                             |
|------------------------------|-------------------------------------------|
| (0, 0, 0, 0)                 | $2 \cdot 10^{4L+3}$                       |
| (0, 0, 1, 0)                 | $27 \cdot 10^{4L+3} - 27 \cdot 10^{3L+3}$ |
| (1, 1, 0, 1)                 | $2 \cdot 10^{4L}$                         |
| (1, 1, 1, 1)                 | $27 \cdot 10^{4L}$                        |
| (0, 1, 0, 1)                 | $2 \cdot 10^{L+3}$                        |
| (0, 1, 1, 0)                 | $27 \cdot 10^{L+3}$                       |

Equivalently, with  $\delta = 10^{-L}$ , the weights are proportional to 2,  $27(1-\delta)$ ,  $\frac{2}{1000}$ ,  $\frac{27}{1000}$ ,  $2\delta^3$ ,  $27\delta^3$  on the same atoms.

**Lemma 4** (Two exact zeros). *For every  $L$ :  $I(A ; B|D) = 0$  and  $I(A ; C|B) = 0$  on  $W_L$ , exactly.*

*Proof.*  $I(A ; B|D)$ : on the three atoms with  $D = 0$  the bit  $A$  is constant ( $= 0$ ), and on the three atoms with  $D = 1$  the bit  $B$  is constant ( $= 1$ ). Conditioned on either value of  $D$ , one of the two arguments of the mutual information is deterministic, so each conditional term vanishes.

$I(A ; C|B)$ : on the two atoms with  $B = 0$  the bit  $A$  is constant ( $= 0$ ), so that block contributes zero. On the  $B = 1$  block the joint distribution of  $(A, C)$  has the  $2 \times 2$  weight matrix  $\begin{pmatrix} w_{0101} & w_{0110} \\ w_{1101} & w_{1111} \end{pmatrix}$ , and  $A$  and  $C$  are conditionally independent iff its determinant vanishes:  $w_{1101} w_{0110} = w_{1111} w_{0101}$ . Both products equal  $54 \cdot 10^{5L+3}$ .  $\square$

Because of Lemma 4, on the family the left side of (1) collapses to

$$f_k(W_L) = k_2 y_L + e_L, \quad y_L := I(C ; B|A), \quad e_L := I(C ; D) - I(C ; AB), \quad (2)$$

for every  $(k_1, k_3)$ : those two coefficients multiply exact zeros.

## 3 Why the ratio diverges

Write  $\delta = 10^{-L}$  and use the normalized form of Definition 3. Two structural observations drive everything.

**Proposition 5.** *As  $\delta \rightarrow 0$ :  $y_L = \Theta(\delta^5)$  while  $-e_L = \Theta(\delta^4)$ , with  $e_L < 0$  for all small  $\delta$ . Consequently  $R_L = -e_L/y_L = \Theta(1/\delta) \rightarrow \infty$ .*

*Proof sketch, elementary and self-contained.* (i)  $y_L$  needs two rare events at once. Only the  $A=0$  block contributes to  $I(C ; B|A)$  (given  $A=1$ ,  $B \equiv 1$ ). Within  $A=0$ , the branch  $B=1$  carries probability  $\Theta(\delta^3)$  (the two  $\delta^3$ -atoms), and conditioned on the two branches the distributions of  $C$  are  $(2, 27)/29$  versus  $(2, 27(1-\delta))/(29-27\delta)$  — total-variation distance  $\Theta(\delta)$ . A mutual information with a  $\Theta(\delta^3)$ -rare branch whose conditional laws differ by  $\Theta(\delta)$  is  $\Theta(\delta^3) \cdot \Theta(\delta^2) = \Theta(\delta^5)$ , by the local  $\chi^2$ -expansion of Kullback–Leibler divergence.

(ii)  $e_L$  vanishes exactly when the  $\delta^3$ -atoms are removed. Set the two  $\delta^3$ -weights to zero. Then only four atoms remain, on which the partition of the sample space induced by  $D$  coincides with the partition induced by  $(A, B)$ : the  $D=0$  atoms are exactly the  $AB=00$  atoms and the  $D=1$  atoms are exactly the  $AB=11$  atoms. Hence  $I(C ; D) = I(C ; AB)$  identically in  $\delta$  on the truncated family, i.e.  $e \equiv 0$ . Restoring the two atoms perturbs each side of  $e_L$  by  $\Theta(\delta^3) \cdot \Theta(\delta) = \Theta(\delta^4)$ : each  $\delta^3$ -atom sits in a  $D$ -block whose conditional law of  $C$  it shifts at relative scale  $\Theta(\delta)$  — the atom  $(0, 1, 1, 0)$  joins the  $D=0$  block, where  $C$ 's law is  $\delta$ -tilted, and the atom  $(0, 1, 0, 1)$  joins the  $D=1$  block, where it is the sole  $\delta$ -scale disturbance. The two placements enter  $I(C ; D)$  and  $I(C ; AB)$  asymmetrically (under  $(A, B)$  both atoms sit in their own block  $AB=01$ , where together they leave  $C$ 's conditional law *exactly* equal to  $(2, 27)/29$ ), and the  $\Theta(\delta^4)$  mismatch has a strictly negative sign for small  $\delta$ , which the interval certificates of Section 4 pin down together with its constant.

Combining (i) and (ii):  $R_L = \Theta(\delta^4)/\Theta(\delta^5) = \Theta(1/\delta)$ .  $\square$

*Remark 6.* The certified computation below measures the sharp law:  $R_L \delta \rightarrow 2$ , i.e.  $R_L = 2 \cdot 10^L - c + o(1)$  with  $c \approx 0.6437$ ; and the per-step ratio  $R_{L+1}/R_L$  is certified  $\geq 1.9$  (measured:  $\rightarrow 10$ ) at every step  $2 \leq L \leq 15$ . Theorem 1 needs only  $R_L \rightarrow \infty$ .

## 4 Machine certificates

The published verifier reproduces, from the integer weights alone, with directed-rounding decimal interval arithmetic (every logarithm bracketed one-sidedly; every certified sign a theorem about the printed digits):

| $L$ | certified lower bound on $R_L$ |
|-----|--------------------------------|
| 2   | 189.4773957                    |
| 3   | 1,998.356506                   |
| 4   | 19,999.25622                   |
| 8   | 199,999,999.3                  |
| 12  | $1.999999999 \cdot 10^{12}$    |
| 16  | $1.999999999 \cdot 10^{16}$    |

together with  $y_L > 0$  and  $e_L < 0$  for  $2 \leq L \leq 16$ , the per-step growth bound, and explicit kill certificates  $f_k(W_L) < 0$  including

$$k = (1, 1, 1) \text{ at } L=2, \quad k = (10^9, 10^9, 10^9) \text{ at } L=15, \quad k = (10^{30}, 100, 10^{30}) \text{ at } L=5,$$

$$k = (0, 10^{20}, 0) \text{ at } L=22, \quad k = (0, 10^{24}, 0) \text{ at } L=26.$$

By (2) a certificate at  $k_2$  covers all  $(k_1, k_2, k_3)$ , so every conjectured constant triple with  $k_2 \leq 10^{24}$  is refuted by a named, explicit distribution; the verifier accepts an arbitrary triple and searches out the certifying  $L$ . Four independently written implementations (interval engines in independent codebases, no shared code) agree on every row.

## 5 Discussion

What survives, and what closes. The Linden–Winter constrained inequality (their Theorem 1) is untouched, as are the infinite constrained families of Cadney–Linden–Winter [5] and Majenz’s partial lifting on his slice [8]. What closes is the named route from the constrained inequality to an unconstrained one: no linear relaxation of the constraints works, through states as simple as six-atom classical distributions on four bits.

Linden and Winter motivated the conjecture by analogy: Zhang and Yeung also first found a constrained Shannon inequality and “only somewhat later their unconstrained inequality” [3]. Completed, the analogy points the other way. Kaced and Romashchenko [7] proved that the Zhang–Yeung 1997 conditional inequality itself does *not* lift — the unconstrained 1998 inequality is a different functional — and the present result extends the pattern to the von Neumann side. For the same reason, refuting Conjecture 6 does not settle whether *some* unconstrained non-von-Neumann-type inequality exists: the Zhang–Yeung precedent shows the lift can fail while a different unconstrained inequality exists. What the result does is remove the one standing candidate, consistent with the current assessment that no such inequality is in sight [6].

Independent checking is deliberately easy: the states are classical, the weights are integers, and the published verifier is dependency-free. The entropy vector of any  $W_L$  can also be fed to the qITIP linear-programming prover of Huang, Rippchen and Berta [9] as a further cross-check.

## References

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